

## RESEARCH ARTICLE OPEN ACCESS

## School Experiences of Gifted Adolescents and Their Peers

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## ABSTRACT

This study examined the school experiences of gifted adolescents and their non-identified peers. Cross-sectional data were obtained from 748 middle and high school students from a large public school district in the Southeastern United States. Students reported on their behavioral and emotional engagement and disaffection, relationships with teachers and peers, social adjustment, and sense of belonging. Gifted students also responded to questions about their perceived social stigma related to giftedness. No multivariate or univariate group differences were found between gifted and non-identified students on the study measures. However, not all gifted students had similar school experiences, particularly in relation to feelings of social stigma. About one-third of the gifted students reported experiencing some degree of stigma. These students had lower teacher-student relationships ( $p < 0.001$ , Cohen's  $d = 0.84$ ), weaker peer support ( $p < 0.01$ , Cohen's  $d = 0.63$ ), lower social adjustment ( $p < 0.01$ , Cohen's  $d = 0.58$ ), and a lower sense of belonging ( $p < 0.001$ , Cohen's  $d = 0.10$ ) compared to other gifted students who experienced little or no stigma. Findings are discussed in the context of gifted education and their implications for educational practice.

The prevalence of psychological distress such as depression, anxiety, and social isolation among school-age adolescents is shockingly high (Bitsko et al. 2018). Data from the Centers for Disease Control and Prevention's (Centers for Disease Control and Prevention 2021) Youth Risk Behavior Survey for 2021 show that 42% of students experienced persistent feelings of sadness or hopelessness almost every day for at least 2 weeks in a row and nearly one-third (29%) reported poor mental health over the past 30 days. The average age of the first onset of depression was about 30 some fifty years ago (Seligman 2011); however, it now ravages adolescents at increasingly younger ages. While there is no clear and convincing evidence on whether these mental-health challenges are more or less common among gifted students, we have good reason to explore ways to better support their school experiences. The fit of individual student to the learning environment, positive school engagement, and supportive teacher-student and peer relationships may serve as protective factors.

Gifted students may have more positive experiences in school when compared to their non-identified peers (Kitsantas et al. 2017; Košir et al. 2015). Their advanced cognitive abilities often translate

into enhanced academic prospects and achievements, more inviting learning environments, and increased opportunities to demonstrate independence (Kitsantas et al. 2017; Vogl and Preckel 2014). Feelings related to academic competence and success can lead to greater interest in school and more rewarding relationships with teachers (Rinn et al. 2011; Vogl and Preckel 2014). Despite the potential for favorable experiences, being academically advanced does not come without adversities. The expectation to excel may cause stress for gifted students (Coleman et al. 2015; Kitsantas et al. 2017; Peterson et al. 2009). They may often feel negative pressure related to academic competition or comparisons, which can also lead to psychological distress (Coleman et al. 2015; Kitsantas et al. 2017; Peterson et al. 2009). Additionally, gifted students may experience feelings of stigma by their high ability (J. R. Cross et al. 2015, 2019), and they may not always be as emotionally adjusted as they appear to be (Olszewski-Kubilius et al. 2014; Peterson et al. 2009).

The purpose of this paper was to explore the similarities and differences in school experiences between gifted adolescents and their counterparts. The analysis draws from a cross-

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**Summary**

- School experiences of gifted students may not differ substantially from those of other students.
- Some gifted students may experience social stigma.
- Social stigma of giftedness is associated with weaker relationships with teachers and peers, lower social adjustment, and a reduced sense of belonging.

sectional study of a large public school district in the state of Georgia that aims to understand students' emotional and behavioral engagement, teacher-student relationships, peer support, sense of belonging, and social adjustment. We also examined perceived social stigma among gifted students.

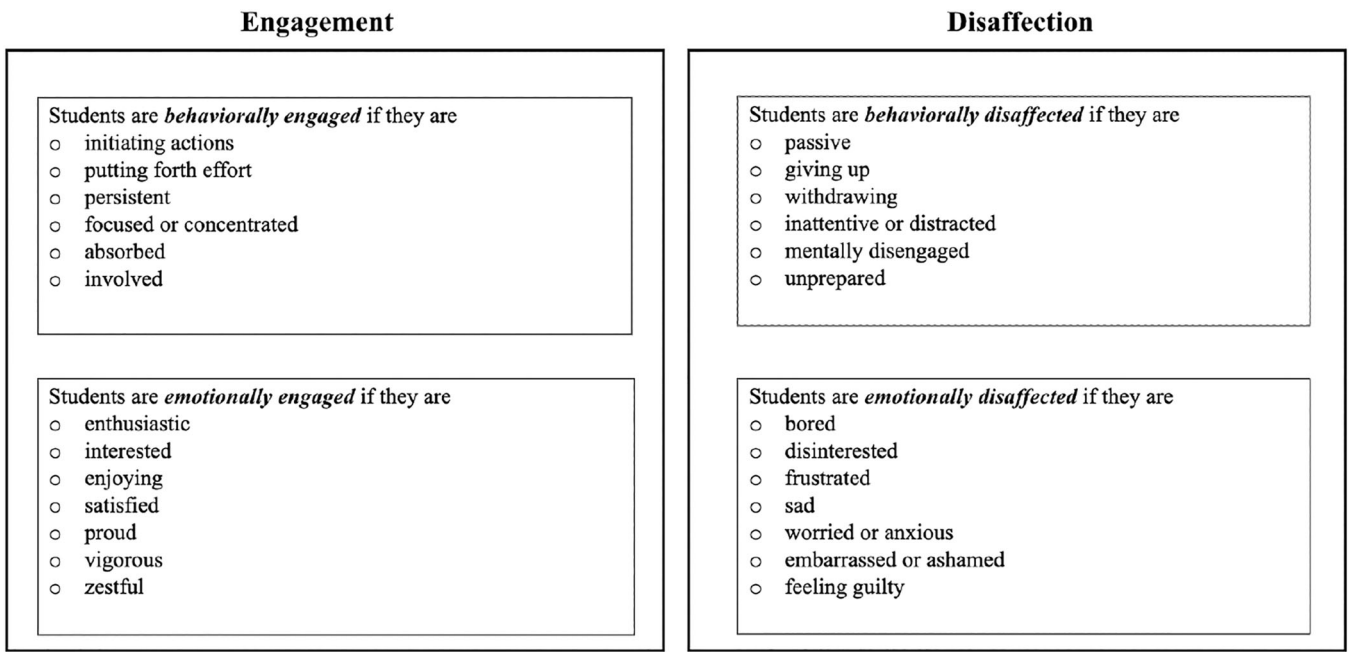
**1 | Student Engagement and Disaffection**

Student engagement (also known as *school engagement*) refers to the quality of students' connection to or involvement with school and school activities (Fredricks et al. 2004; Skinner et al. 2009). There is strong consensus that student engagement is a meta-construct that is both multifaceted and malleable (Wong and Liem 2022). The Self-System Model of Motivational Development (SSMMD; Skinner et al. 2009) is one of the prevailing models of student engagement. According to this model, engagement is "the outward manifestation of a motivated student" (Skinner et al. 2009, p. 494) and could manifest in both behavioral and emotional forms. Skinner et al. (2008) proposed four types of engagement components: behavioral engagement, emotional engagement, behavioral disaffection, and emotional disaffection (see Figure 1).

SSMMD draws on self-determination theory (SDT; Deci and Ryan 2000) and explains engagement in terms of the

satisfaction of three fundamental psychological needs: autonomy (i.e., the need to have ownership of one's actions), competence (i.e., the need to experience mastery), and relatedness (i.e., the need to feel connected to others). When these needs are met, engagement occurs and manifests itself in behaviors and emotions (Connell and Wellborn 1991). In a school context, behavioral engagement refers to students' active and voluntary participation in the class, such as effort and persistence. Emotional engagement, on the other hand, includes more implicit factors, such as belonging, enthusiasm, and a positive attitude toward learning (Appleton et al. 2006; Fredricks et al. 2004; Skinner et al. 2009). These two engagement types are closely linked to each other despite not directly affecting one another (Skinner et al. 2008). When psychological needs are not met, students may experience disaffection. Skinner et al. (2008) argue that disaffection is not merely the absence of engagement. Instead, it includes maladaptive motivational states in which students lack drive and act out of external pressure rather than internal motivation. In a later study, Skinner et al. (2009) further distinguished disaffection from engagement by noting that disaffection includes behaviors such as passivity, lack of initiative, mental withdrawal, lack of effort and attention, and giving up.

The lived experiences of gifted students at school can be diverse, complex, and sometimes negative, similar to those of their non-identified peers (T. L. Cross et al. 2018). Gifted students often demonstrate asynchronous development, where their cognitive abilities outpace their physical, emotional, or social development (Silverman 2012). Small differences in the early years can grow into widened gaps both within the individual student and between the student and age-norm peers. This asynchrony can lead to increased challenges in balancing uneven development in everyday life and interactions, which may significantly impact their engagement in school. In addition, gifted students' high academic abilities can mask difficulties they experience in other developmental domains. As Moon (2009) noted, "high-



**FIGURE 1** | A Motivational Conceptualization of Engagement and Disaffection. *Note:* Adapted from Skinner et al. (2008).

ability students in unchallenging educational environments may look problem-free because they are performing above grade level, when, in fact, they are developing maladaptive motivational beliefs that will sabotage their resilience when they encounter more challenging coursework in the future.” (p.274).

Gifted students typically report higher levels of motivation for school (Gottfried and Gottfried 1996; Hornstra et al. 2020; Wirthwein et al. 2019). However, while they enjoy school, they often lose interest in schoolwork when it is not challenging enough. Kanevsky and Keighley (2003) interviewed a group of gifted high school students they called “nonproducers,” who had gradually disengaged from the school experience. The lack of challenge was one of the core themes that students attributed to their increasing behavioral and emotional disaffection. In a recent 2-year longitudinal study, Hornstra et al. (2023) examined academic motivation of gifted students and their classmates in mixed-ability classrooms. Although gifted students initially reported more favorable motivation and better engagement, they lost this advantage over time. It is not surprising that gifted students may be at risk for developing unfavorable motivational patterns or becoming noticeably disengaged when “... their advanced development comes into contact with uninteresting, undemanding, and slow-moving curriculum” (Coleman et al. 2021, p. 445). In most cases, school experiences for gifted learners do not provide appropriate levels of challenge to help them move on to more advanced work (Coleman et al. 2021; Little 2012; Rogers 2007). This mismatch between student ability and provided programs may limit engagement and result in academic resistance, underachievement, disconnect from school, and even sometimes being bullied (Coleman et al. 2015).

## 2 | Psychological Engagement: Sense of Belonging and Relationships

Student disengagement is not an instantaneous event, but rather a gradual process that occurs over time. Although a motivational perspective can help identify a student's level of engagement, its intensity, and its quality (Maehr and Meyer 1997), it is not sufficient on its own to capture the gradual process by which students become disengaged in learning. To better understand this process—and be able to implement timely interventions to prevent students from becoming further disengaged from school and learning - we must consider the goodness of fit between students and their learning environments and focus on the alterable factors which address underlying psychological processes. For the present study, we chose to examine another type of engagement, *psychological engagement* (Appleton et al. 2006), which includes less observable indicators such as feelings of belonging and relationships with teachers and peers.

### 2.1 | Sense of School Belonging

Sense of school belonging refers to the extent to which students feel respected, are comfortable, and supported by others in the school community (Anderman 2003). Students' degree of belonging has been associated with adaptive patterns of motivation (Sánchez et al. 2005), positive school-related affect

(Raufelder et al. 2015), and academic success (Faircloth and Hamm 2005). To students who are identified as gifted, school belonging and feelings of social support can be pivotal. Research findings are mixed as to whether gifted students experience higher or lower levels of school belonging compared to their peers. Some studies suggest that gifted students may feel a greater sense of belonging (Cash and Lin 2022; Vogl and Preckel 2014). Vogl and Preckel (2014), for example, noted that the advanced cognitive skills of gifted students may enable them to be more perceptive in social situations, thus aiding them in interpersonal encounters. Additionally, their advanced thought processes may improve gifted students' interactions with teachers, thereby supporting their sense of belonging (Rinn et al. 2011; Vogl and Preckel 2014). However, other research indicates that gifted students may face unique challenges that can weaken their sense of belonging. Some gifted students may experience feelings of rejection even when their peers offer acceptance (Košir et al. 2015; Lee et al. 2012; Peterson et al. 2009; Schmitt and Goebel 2015). These perceptions may be self-induced by gifted students who believe they cannot be authentic at school due to insecurities related to their differences (Coleman et al. 2015; Rinn et al. 2011). Some gifted students may attribute their differences to divergent interests and passions instead of their academic abilities (Bailey 2011; Coleman et al. 2015). Several studies proposed that gifted students may believe their non-gifted peers and educators view their giftedness unfavorably (Olszewski-Kubilius et al. 2014; Schmitt and Goebel 2015).

### 2.2 | The Social Stigma of Giftedness

Being identified as gifted may limit a student's sense of acceptance among peers and, therefore, lead them to feel stereotyped and restricted in their social interactions (T. L. Cross et al. 2018). This phenomenon, referred to as “the stigma of giftedness,” has been explored through Coleman (1985) Stigma of Giftedness Paradigm (SGP). The SGP is built on three basic tenets:

1. Gifted students desire normal social interactions.
2. When their giftedness becomes known to others, gifted students believe that they will be treated differently.
3. Gifted students tend to manage the information they reveal about their abilities to maintain a great amount of social latitude.

Tannenbaum (1983) observed that gifted students are influenced by the way others feel about their abilities. Mendaglio (2012) argued that gifted students feel stigmatized and that this feeling arises directly from the label of being gifted. Multiple studies have shown that gifted students are aware of the SGP and frequently manage the information others have about them to achieve more typical social interactions within the school environment (Coleman and Cross 1988; J. R. Cross et al. 2019; T. L. Cross et al. 1995; T. L. Cross and Stewart 1995). When individuals feel stigmatized, they may experience reductions in the levels of psychological need satisfaction (Williams and Gerber 2005). Therefore, they may attempt to use their cognitive, emotional, and behavioral resources to restore that sense of satisfaction (Mammadov 2021). In line with the SGP, gifted

students may engage in efforts to mask their intellectual differences, particularly those that are less overtly visible (T. L. Cross 1997).

## 2.3 | Teacher-Student Relationships

Two important components of the sense of belonging are positive relationships with teachers and a network of peers through which the student feels recognized (Faircloth and Hamm 2005). Certain social and emotional problems we have previously noted, such as ostracism and disaffection, which gifted students may experience, can often be addressed through rewarding attachments to school and supportive teachers and peer groups. The literature is clear that teacher-student relationships strongly influence many social-emotional and learning outcomes (Crosnoe et al. 2004; Kang et al. 2021; Mammadov and Schroeder 2023; Skinner et al. 2008). Students who experience teacher support and positive relationships are more likely to develop a stronger sense of belonging in school (Furrer and Skinner 2003), show greater intrinsic motivation (Hornstra et al. 2020), achieve higher academic performance (McCormick and O'Connor 2015; Zee et al. 2013), and demonstrate better classroom engagement (O'Connor and McCartney 2007; Roorda et al. 2011; Skinner et al. 2008). Kanevsky and Keighley (2003) identified five interdependent features (5C) that shape gifted students' school experiences: control, choice, challenge, complexity and caring. They noted that gifted students are especially prone to boredom when they lack control and choice, as they often feel they do not receive sufficient autonomy support from teachers. They also emphasized that caring teachers, representing the last of the five Cs, have the power to overcome the other four Cs. A caring teacher, according to students interviewed in this study, is someone who has mutual respect and reciprocity, flexibility in thinking and teaching, and a willingness to listen to student inquiries.

## 2.4 | Peer Support

Peer support is as important for students as having healthy relationships with teachers. Positive peer relations have been linked to greater school engagement and improved academic performance (Appleton et al. 2006; Perdue et al. 2009), whereas difficulties with peers are associated with depression and other social and emotional issues (Buck and Dix 2012). Gifted students are not a homogeneous group, nor are the contexts of previous studies similar enough for us to draw out conclusive insights about the peer relationship experiences of gifted students. Findings are mixed. Some studies have reported no differences in peer relations and social adjustment between gifted students and their non-identified peers (Košir et al. 2015; Neihart 2002). Some other studies suggested that gifted students were more competent both academically and socially (Lee et al. 2012; Mueller 2009). In the context of SGP, one might expect that gifted students could be disadvantaged in peer relations because they experience the stress of being labeled as gifted (T. L. Cross 1997; Pfeiffer 2009). Additionally, as noted earlier, gifted students may experience asynchrony in their development, which may affect their role in peer groups, their

social acceptance, and the support they receive from peers. Research indicates that gifted students believe that their differences in certain areas impact their relationships with others (Gross 2004). As children mature, their social circles get wider and get less supervised by adults (Rubin et al. 2006). Social issues and relations start to matter more (T. L. Cross et al. 2018), and students become more concerned about 'fitting in' with their peer groups (Rubin et al. 2006). We should therefore prioritize identifying contexts and systems to foster the development of social competence skills for all students, including gifted students, with the hope of minimizing the risk of maladaptation in their later life.

## 2.5 | Purpose

We examined differences in school experiences between gifted students and their non-identified peers. Specifically, we aimed to understand if gifted students experience behavioral and emotional engagement, sense of belonging, and social adjustment differently, and whether they perceive their relationships with teachers and peer support differently. We also examined perceived social stigma among gifted students due to their giftedness. That is, we did not expect all gifted students to feel they are treated differently, but some may. Therefore, we examined how the SGP applies to these students and whether those experiencing some degree of stigma differ in their engagement, sense of belonging, social adjustment, teacher-student relationships, and perceived peer support. To guide our analysis, we addressed the following research questions:

1. Do gifted students report statistically different levels of engagement and disaffection (behavioral and emotional), social adjustment, teacher-student relationships, peer support, and sense of belonging compared to their non-identified peers?
2. How is the social stigma of giftedness related to gifted students' school experiences?

## 3 | Methodology

### 3.1 | Participants

Participants included 748 middle and high school students from a large public school district in the Southeastern United States. Students were recruited from four middle schools ( $n = 551$ , 74%) and two high schools ( $n = 193$ , 26%). The sample consisted of 334 girls (45%) and 286 boys (38%), with 128 students (13%) not reporting their gender. The majority of the students were White (65%), followed by Black (6%), Hispanic (5%), and Biracial/Multiracial (4%). A total of 257 students (34%) reported being enrolled in their school's gifted program and services. A more detailed demographic breakdown of the participants is presented in Table 1.

Students who receive gifted services were identified in elementary or middle school according to Georgia's multiple eligibility criteria. Students' mental ability, achievement, creativity, and motivation were evaluated and scored based on national norms as outlined in the Georgia Department of



**TABLE 1** | Participant demographics.

| Demographics                        | N   | %     |
|-------------------------------------|-----|-------|
| Gender                              |     |       |
| Male                                | 286 | 38.24 |
| Female                              | 334 | 44.65 |
| Prefer not to answer/Not reported   | 128 | 17.11 |
| Giftedness                          |     |       |
| Gifted                              | 257 | 34.36 |
| Non-Identified                      | 385 | 51.47 |
| Not reported                        | 106 | 14.17 |
| Grade Level                         |     |       |
| Middle School                       | 551 | 73.66 |
| High School                         | 193 | 25.80 |
| Not reported                        | 4   | 0.53  |
| Ethnicity                           |     |       |
| American Indian/Alaska Native       | 16  | 2.14  |
| Asian                               | 7   | 0.94  |
| Biracial/Multicultural              | 28  | 3.74  |
| Black/African American              | 43  | 5.75  |
| Hispanic/Latinx                     | 38  | 5.08  |
| Native Hawaiian or Pacific Islander | 3   | 0.40  |
| White                               | 484 | 64.71 |
| Other                               | 26  | 3.48  |
| Total                               | 748 | 100   |

Education (2024) *Resource Manual for Gifted Education Services*. Once students were identified for gifted placement, they continued to maintain this status throughout their schooling. Though any student in high school can be placed into advanced, dual enrollment, or AP classes, students are only assigned a gifted label if they were formally tested before high school. The non-identified students were students receiving regular education without any special education services or accommodations. They were selected from the same schools as the gifted students and they were comparable to the gifted students in terms of key demographics such as gender, ethnicity, and grade level.

### 3.2 | Procedure

The study commenced with a formal request for collaboration sent to the superintendent of the district, seeking consent. Subsequently, ethical approval was obtained from the University of Georgia Institutional Review Board. Following this, the school administrations contacted parents to inform them about the study and obtain their consent. Only students who had parental consent were recruited for the study. Students were also informed about the research and their assent was collected. Students anonymously and voluntarily participated in the study during school hours. There were no personal identifiers in the study and authors did not have access to any information that could lead to the data being linked to specific students.

### 3.3 | Measures

#### 3.3.1 | Behavioral and Emotional Engagement

Eighteen items from the Engagement Versus Disaffection with Learning: Student Report (Skinner et al. 2009) were used to measure students' behavioral engagement ( $\alpha = 0.76$ ; e.g., "I try hard to do well in school."), emotional engagement ( $\alpha = 0.82$ ; e.g., "When I'm in class, I feel good."), behavioral disaffection ( $\alpha = 0.81$ ; e.g., "When I'm in class, my mind wanders."), and emotional disaffection ( $\alpha = 0.78$ ; e.g., "When I'm doing work in class, I feel bored."). Items were rated on a 4-point Likert scale ranging from *not at all true* (1) to *very true* (4). Higher scores indicate greater engagement on the engagement subscales and greater disaffection on the disaffection subscales. The scale has demonstrated good reliability and construct validity (Skinner et al. 2009) and has been widely used in previous research (Fredricks and McColskey 2012).

#### 3.3.2 | Affective Engagement: Teacher-Student Relationships and Peer Support

The Student Engagement Instrument (Appleton et al. 2006) was used to assess students' affective engagement regarding their relationships with teachers and peers. The teacher-student relationship subscale includes 9 items ( $\alpha = 0.92$ ; e.g., "My teachers are there for me when I need them."), whereas the peer support subscale has 6 items ( $\alpha = 0.90$ ; e.g., "Other students here like me the way I am."). This 15-item measure used a 5-point Likert-type scale, ranging from *strongly disagree* (1) to *strongly agree* (5), with higher scores indicating stronger relationships with teachers and peers. The scale was originally constructed with a diverse sample (Appleton et al. 2006) and it has been strengthened by further validation studies (Betts et al. 2010; Lovelace et al. 2014; Reschly et al. 2014).

#### 3.3.3 | Sense of Belonging

The Psychological Sense of School Membership Scale (Goodenow 1993) it was used to assess students' sense of belonging at their current school ( $\alpha = 0.93$ ; e.g., "I feel like a real part of my school."). Five negatively worded items (e.g., "It is hard for people like me to be accepted here.") were reverse coded before data analysis to ensure consistent interpretation. The scale includes 18 items rated on a 5-point Likert-type scale, ranging from *not at all true* (1) to *completely true* (5). A composite score was calculated, with higher scores reflecting a stronger sense of belonging at school. The scale has demonstrated strong psychometric properties and is widely recognized as a reliable measure across diverse student populations (Korpershoek et al. 2020).

#### 3.3.4 | Social Adjustment

The social adjustment subscale of Adolescent Multidimensional Social Competence Questionnaire (Gómez-Ortiz et al. 2017) was used in this study. The subscale consists of eight items ( $\alpha = 0.88$ ; e.g., "I join in with the activities that others take part in.") rated on a 5-point Likert scale, ranging from *never* (1) to *always* (5).

Higher scores indicate a greater level of social adjustment. The scale has been cross-culturally validated in diverse countries and has demonstrated strong psychometric properties consistent with the original version (e.g., Dinh et al. 2021; Gómez-Ortiz et al. 2019; Twardowska-Staszek and Zych 2023; Zych et al. 2018).

### 3.3.5 | Perceived Stigma

Gifted students completed four items, each regarding their perceived social stigma related to giftedness. The items were: (1) “I feel different from other students because I am gifted”, (2) “I feel isolated or lonely because I am gifted”, (3) “My teacher treats me differently because I am gifted”, and (4) “Other students treat me differently because I am gifted”. Each item was rated on a 4-point Likert scale, ranging from *never* (1) to *often* (4). The scale demonstrated good internal consistency, with a Cronbach’s alpha of 0.80.

## 4 | Data Analysis

Data were analyzed in R Studio (version 4.2.1). We used Full Information Maximum Likelihood (FIML) to estimate missing data. The percentage of missing data was 4.34%. Our assessment of missing data pattern did not provide sufficient evidence to reject the assumption of data being Missing Completely at Random (MCAR), supporting the use of FIML for handling the missing data. To examine the differences between gifted and non-identified students across study measures, we employed Multivariate Analysis of Covariance (MANCOVA). Gender, ethnicity, and grade level (middle vs. high school) as covariates. We also included the interaction between gifted identification and grade level in our model. We planned to conduct follow up Univariate ANCOVAs to assess group differences on individual constructs, regardless of the overall significance of the MANCOVA results. Bonferroni corrections were applied to control for type I error when running univariate analyzes. Finally, we classified gifted students by their perceived stigma levels and compared these groups on school engagement, teacher and peer relationships, social adjustment, and sense of belonging.

## 5 | Results

Descriptive statistics for the types of school engagement and social adjustment are presented in Table 2. On average, students reported relatively high levels of behavioral engagement ( $M = 3.35$ ,  $SD = 0.47$ ) and moderate levels of emotional engagement ( $M = 2.86$ ,  $SD = 0.61$ ). Similarly, students experienced slightly higher emotional disaffection ( $M = 2.51$ ,  $SD = 0.73$ ) compared to behavioral disaffection ( $M = 2.29$ ,  $SD = 0.66$ ). Teacher-student relationships ( $M = 3.57$ ,  $SD = 0.82$ ) and peer support ( $M = 3.54$ ,  $SD = 0.83$ ) were rated fairly high, as well as sense of belonging ( $M = 3.50$ ,  $SD = 0.76$ ). Social adjustment had the highest mean score ( $M = 3.65$ ,  $SD = 0.76$ ), suggesting that students felt comfortable in adapting socially within the school environment. When looking at Table 2, we observed no remarkable differences between gifted and other students. Zero-order correlations among

TABLE 2 | Descriptive statistics for study variables.

|                              | Total<br>M (SD) | Gifted<br>M (SD) | Non-identified<br>M (SD) | Middle School<br>M (SD) | High School<br>M (SD) | Female<br>M (SD) | Male<br>M (SD) |
|------------------------------|-----------------|------------------|--------------------------|-------------------------|-----------------------|------------------|----------------|
| Behavioral engagement        | 3.35 (0.47)     | 3.35 (0.48)      | 3.34 (0.46)              | 3.34 (0.45)             | 3.36 (0.50)           | 3.42 (0.40)      | 3.28 (0.52)    |
| Emotional engagement         | 2.86 (0.61)     | 2.88 (0.61)      | 2.86 (0.61)              | 2.85 (0.63)             | 2.90 (0.55)           | 2.87 (0.57)      | 2.87 (0.67)    |
| Behavioral disaffection      | 2.29 (0.66)     | 2.29 (0.69)      | 2.29 (0.64)              | 2.26 (0.65)             | 2.40 (0.68)           | 2.24 (0.60)      | 2.32 (0.69)    |
| Emotional disaffection       | 2.51 (0.73)     | 2.51 (0.77)      | 2.49 (0.70)              | 2.48 (0.74)             | 2.59 (0.70)           | 2.52 (0.71)      | 2.47 (0.74)    |
| Teacher-student relationship | 3.57 (0.82)     | 3.56 (0.83)      | 3.61 (0.81)              | 3.58 (0.84)             | 3.59 (0.75)           | 3.60 (0.77)      | 3.62 (0.85)    |
| Peer support                 | 3.54 (0.83)     | 3.54 (0.83)      | 3.55 (0.84)              | 3.54 (0.83)             | 3.55 (0.85)           | 3.50 (0.83)      | 3.66 (0.82)    |
| Social adjustment            | 3.65 (0.76)     | 3.69 (0.74)      | 3.63 (0.78)              | 3.64 (0.76)             | 3.68 (0.76)           | 3.73 (0.71)      | 3.60 (0.81)    |
| Sense of belonging           | 3.50 (0.76)     | 3.49 (0.77)      | 3.54 (0.76)              | 3.53 (0.77)             | 3.46 (0.74)           | 3.51 (0.75)      | 3.56 (0.75)    |
| Stigma*                      | NA              | 1.97 (0.82)      | NA                       | 1.93 (0.83)             | 2.02 (0.81)           | 1.99 (0.85)      | 1.98 (0.80)    |

\*Perceived Social Stigma was measured among gifted students only.

**TABLE 3** | Zero-Order Correlations.

| Study variables                 | 1     | 2     | 3     | 4     | 5    | 6    | 7    | 8 |
|---------------------------------|-------|-------|-------|-------|------|------|------|---|
| 1. Behavioral engagement        |       |       |       |       |      |      |      |   |
| 2. Emotional engagement         | 0.60  |       |       |       |      |      |      |   |
| 3. Behavioral disaffection      | −0.60 | −0.54 |       |       |      |      |      |   |
| 4. Emotional disaffection       | −0.42 | −0.73 | 0.62  |       |      |      |      |   |
| 5. Teacher-student relationship | 0.38  | 0.61  | −0.41 | −0.56 |      |      |      |   |
| 6. Peer support                 | 0.28  | 0.47  | −0.29 | −0.40 | 0.57 |      |      |   |
| 7. Social adjustment            | 0.38  | 0.50  | −0.33 | −0.39 | 0.52 | 0.73 |      |   |
| 8. Sense of belonging           | 0.38  | 0.62  | −0.46 | −0.59 | 0.77 | 0.77 | 0.76 |   |

Note: All correlations were significant at  $p < 0.001$ .

the study variables are presented in Table 3. Both behavioral and emotional disaffection were moderately and negatively correlated with social adjustment and sense of belonging. Peer support had a strong positive correlation with social adjustment, whereas teacher-student relationship showed a moderate positive correlation. Both teacher-student relationship and peer support showed strong associations with sense of belonging.

The MANCOVA showed no overall significant multivariate effect of gifted identification on the combined dependent variables (Pillai's Trace = 0.010,  $F(8, 622) = 0.783$ ,  $p = 0.617$ ). In addition, the interaction between gifted identification and grade level (middle school vs. high school) was also not significant (Pillai's Trace = 0.020,  $F(8, 622) = 1.425$ ,  $p = 0.184$ ), suggesting that differences between gifted students and their non-identified counterparts do not vary across grade levels. Univariate ANCOVAs, adjusted for gender, ethnicity, and grade level, using Bonferroni corrections did not reveal any meaningful differences for individual constructs, as all  $p$ -values were nonsignificant. Gifted students did not differ from their peers in emotional and behavioral engagement and disaffection, perceived teacher-student relationship, peer support, sense of belonging, or social adjustment.

Gifted students in our sample responded to items related to perceived social stigma of giftedness, which yielded a mean score of 1.98 (SD = 0.82). We categorized perceived stigma levels into three groups: Students whose  $z$ -scores were 1.5 or more standard deviations above the mean were classified as experiencing high stigma, those scoring between 0.5 and 1.5 standard deviations above the mean were considered to have low to moderate stigma, and students scoring below 0.5 standard deviation above the mean were categorized as having relatively no stigma. The distribution of responses suggested that 12% of students ( $N = 30$ ) perceived high stigma, 19% reported low to moderate stigma ( $N = 49$ ), and the remaining 69% experienced relatively no stigma ( $N = 177$ ).

The sample sizes of the high and low to moderate stigma groups were small. We combined them into a single "Stigma" group and compared it to the "No Stigma" group across behavioral engagement, emotional engagement, behavioral disaffection, emotional disaffection, teacher-student relationship, peer support, social adjustment, and sense of belonging. A series of independent-samples  $t$ -tests with 1000 bootstrap resampling

was conducted using Bonferroni correction to account for multiple comparisons, where each  $p$ -value was multiplied by the total number of comparisons (8). The Stigma group was found to have lower teacher-student relationships ( $p < 0.001$ ), weaker peer support ( $p < 0.01$ ), lower social adjustment ( $p < 0.01$ ), and a lower sense of belonging ( $p < 0.001$ ). Table 4 presents  $t$ -test results.

## 6 | Discussion

The most important finding from this study is that gifted students did not differ from their peers in their overall school experiences. Students specifically reported high levels of behavioral engagement, teacher-student relationships, perceived peer support, sense of belonging, and social adjustment. No significant differences were observed between gifted and non-identified students. The close mean scores across all comparison measures and high scores on specific ones, actually represent a positive finding, because they challenge the *dis-harmony hypothesis* – the idea that giftedness is associated with the risk for socioemotional downsides and adjustment difficulties (Baudson and Preckel 2013; Mammadov 2023). These results are largely consistent with previous research suggesting that gifted students are no more or less likely to be vulnerable to social and emotional difficulties than other students (Neihart et al. 2015; Shechtman and Silektor 2012). At the very least, gifted students do not appear to experience greater challenges in engagement, social adjustment, or relationships. Rinn (2018) suggested that such difficulties may arise when there is a mismatch between a student and their environment. In our findings, the school environment clearly plays a critical role in supporting positive engagement experiences. It is likely that, for most students, regular instructional practices and the content of gifted programming in participating schools match with a range of student abilities and provide them with appropriate instructional challenges.

The finding that, overall, students reported high levels of engagement should be interpreted cautiously. When considering these results related to engagement and disaffection, it is important to account for potential limitations associated with self-report data, which may be susceptible to social desirability bias (Reschly and Christenson 2012). Students may overstate positive behaviors, such as engagement, and underreport

**TABLE 4** | *t*-test results comparing the Stigma group and other gifted students.

| Study Variables              | Mean | <i>t</i> | <i>p</i> | Bootstrap SE | Adjusted <i>p</i> | Cohen's <i>d</i> |
|------------------------------|------|----------|----------|--------------|-------------------|------------------|
| Behavioral engagement        |      |          |          |              |                   |                  |
| No stigma                    | 3.38 | 1.422    | 0.158    | 0.073        | 1.00              | 0.21             |
| Stigma                       | 3.28 |          |          |              |                   |                  |
| Emotional engagement         |      |          |          |              |                   |                  |
| No stigma                    | 2.96 | 2.909    | < 0.01   | 0.082        | 0.07              | 0.40             |
| Stigma                       | 2.72 |          |          |              |                   |                  |
| Behavioral disaffection      |      |          |          |              |                   |                  |
| No stigma                    | 2.21 | −2.673   | < 0.01   | 0.097        | 0.14              | −.38             |
| Stigma                       | 2.47 |          |          |              |                   |                  |
| Emotional disaffection       |      |          |          |              |                   |                  |
| No stigma                    | 2.43 | −2.342   | < 0.05   | 0.106        | 0.33              | −0.32            |
| Stigma                       | 2.67 |          |          |              |                   |                  |
| Teacher–student relationship |      |          |          |              |                   |                  |
| No stigma                    | 3.76 | 5.569    | < 0.001  | 0.119        | < 0.001           | 0.84             |
| Stigma                       | 3.10 |          |          |              |                   |                  |
| Peer support                 |      |          |          |              |                   |                  |
| No stigma                    | 3.69 | 4.235    | < 0.001  | 0.117        | < 0.01            | 0.63             |
| Stigma                       | 3.19 |          |          |              |                   |                  |
| Social adjustment            |      |          |          |              |                   |                  |
| No stigma                    | 3.82 | 4.091    | < 0.001  | 0.103        | < 0.01            | 0.58             |
| Stigma                       | 3.41 |          |          |              |                   |                  |
| Sense of belonging           |      |          |          |              |                   |                  |
| No stigma                    | 3.70 | 6.685    | < 0.001  | 0.092        | < 0.001           | 0.10             |
| Stigma                       | 3.06 |          |          |              |                   |                  |

negative experiences like disaffection. There may also be a potential source of engagement bias (Reschly et al. 2008), as we might have more students in the sample who were highly engaged and less disaffected. Students with higher levels of school disaffection may have been less likely to participate in the study.

Not all gifted students had similar school experiences, particularly when it came to feeling socially stigmatized due to their giftedness. About one-third of the gifted students reported experiencing some degree of stigma. These students tended to have weaker relationships with teachers and peers, lower social adjustment, and a lower sense of belonging. Our findings align with previous research showing that giftedness can act as a social handicap for some students (J. R. Cross et al. 2015, 2019; Mammadov 2019; Mendaglio 2012). We acknowledge the stigma of giftedness as a fact, as it has been reported in many studies; however, quantitative research on this phenomenon is limited, making it difficult to compare and discuss how common it typically is among gifted students.

Of course, the prevalence of social stigma may vary in different samples from other school districts. We cannot generalize our findings broadly. Just as students' school experiences are shaped by school-related factors—such as available support and the quality of instructional practice—the percentage of gifted

students who feel stigmatized is likely influenced by factors from identification practices to the curricular and programmatic approaches used to support these students. Schools have the potential to create a supportive and inclusive environment that alleviates some possible challenges for gifted students. Even though gifted students have different abilities and needs than their peers, school systems can support these students to an extent where they are not stigmatized, isolated, or singled out due to their diverse abilities. Strategic instructional design, counseling, appropriate educational placement, and effective pedagogical strategies can serve as the primary means to address social and emotional needs of gifted learners (Mammadov and Hertzog 2023).

With several important and interesting findings, there are also new questions that future studies could address. For example, students reported lower emotional than behavioral engagement and experienced greater emotional disaffection than behavioral disaffection. This finding was for the entire sample. When gifted students were considered separately, it held true. We did not explore potential causes of these differences, but our correlation findings suggested that emotional engagement and disaffection were more strongly associated with social and psychological aspects of the school environment, such as teacher-student relationships, peer support, and sense of belonging. These contextual factors, and perhaps some others as well, may be worth



examining to better understand why emotional and behavioral responses differed, which could inform specific interventions or strategies to address emotional disaffection in particular.

The Stigma group reported significantly weaker teacher-student relationships and perceived peer support compared to other gifted students and their non-identified counterparts. The effect size for teacher-student relationship was large (Cohen's  $d = 0.84$ ), suggesting that students in the Stigma group experienced -or perceived- their relationships with teachers as substantially weaker. Peer support and social adjustment also showed meaningful differences, with medium to large effect sizes (Cohen's  $d = 0.63$  and  $0.58$ , respectively). Although the Stigma group reported lower emotional engagement and higher levels of both behavioral and emotional disaffection, these differences became statistically nonsignificant after the Bonferroni correction. School experiences of this group of gifted students should be brought to the forefront of educators' attention. Kanevsky and Keighley's (2003) 5C framework might be a helpful tool to use in looking at how teachers can best address their feelings of disconnectedness. Teacher-student relationships could be improved through professional developments and trainings geared toward addressing the needs of gifted learners (Bangel et al. 2010; Geake and Gross 2008). Weaker peer support, weaker social adjustment, and a lower sense of belonging in the Stigma group could be perceived or founded, both of which align with the literature (Gross 2004; Košir et al. 2015; Lee et al. 2012; Olszewski-Kubilius et al. 2014; Schmitt and Goebel 2015). Further investigation would need to occur to determine the source of these feelings within this particular population, and it is highly likely that the reasons will vary based on individual experiences. Regardless of the basis for these emotions, they should not be ignored as they are a real part of these students' experiences, and they may result in negative perceptions toward school if allowed to persist.

This study, like all other studies, is not free from limitations. First, the sample is drawn from a single school district in the Southeastern United States. Our results are limited to be generalized to the broader gifted population. Future studies could replicate this study in other regions to assess the generalizability of the findings across different cultural and educational contexts. Additionally, we did not explore subgroups within the gifted population (e.g., twice-exceptional students, under-achievers), which would help better understand their unique school experiences. Future research could expand on these findings by examining how these subgroups experience school in terms of their engagement, sense of belonging, social adjustment, and relationships with teachers and peers. Second, disaffected students may be underrepresented in the sample, as they are less likely to participate in voluntary studies such as this one, potentially introducing sampling bias into the study results. Third, we were limited to a small sample size of gifted students compared to other students, which constrained the depth of our analysis. Fourth, our measures were self-report and potentially susceptible to social desirability bias. Future studies should aim to obtain larger sample sizes and include reports from multiple sources. Finally, the study is cross-sectional. Longitudinal studies can help understand more detailed differences between experiences of gifted students and their non-identified peers.

## 7 | Conclusions

A number of conclusions and recommendations can be drawn from the present study. The negative portrayal of gifted individuals in media and popular culture often leads to a disproportionate focus on socioemotional downsides of giftedness, which may cause some to view gifted students as struggling with school engagement and relationships. Our findings suggest that being gifted does not inherently increase the likelihood of such difficulties. Educators need not feel that identifying students as gifted is setting them up for a more difficult school experience. To better support the academic, social, and emotional well-being of gifted students, we recommend the following based on our findings and discussion:

- Provide appropriate instructional challenges that meet the specific needs of the gifted students.
- Engage in professional development designed to help teachers build positive relationships with gifted learners.
- Create opportunities for students to positively engage with and support their peers.
- Address the potential stigma associated with the 'gifted' label, and work to destigmatize thought processes when necessary.
- Be aware of the potential for some gifted students to struggle with social interactions and to develop a sense of belonging and be prepared to offer appropriate support or referrals to specialists as needed.
- Acknowledge the emotional struggles that some gifted students may experience. These challenges are real even if not grounded in objective difficulties and should be met with empathy and support.

The school experiences of all students, including those identified as gifted, can be significantly enhanced by providing an appropriate learning environment, opportunities for positive engagement, and supportive relationships with teachers and peers.

### Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

### References

- Anderman, L. H. 2003. "Academic and Social Perceptions as Predictors of Change in Middle School Students' Sense of School Belonging." *Journal of Experimental Education* 72, no. 1: 5–22. <https://doi.org/10.1080/00220970309600877>.
- Appleton, J. J., S. L. Christenson, D. Kim, and A. L. Reschly. 2006. "Measuring Cognitive and Psychological Engagement: Validation of the Student Engagement Instrument." *Journal of School Psychology* 44, no. 5: 427–445. <https://doi.org/10.1016/j.jsp.2006.04.002>.
- Bailey, C. L. 2011. "An Examination of the Relationships Between Ego Development, Dabrowski's Theory of Positive Disintegration, and the Behavioral Characteristics of Gifted Adolescents." *Gifted Child Quarterly* 55, no. 3: 208–222. Sage Journals. <https://doi.org/10.1177/0016986211412180>.
- Bangel, N. J., S. M. Moon, and B. M. Capobianco. 2010. "Preservice Teachers' Perceptions and Experiences in a Gifted Education Training

- Model." *Gifted Child Quarterly* 54, no. 3: 209–221. Sage Journals. <https://doi.org/10.1177/0016986210369257>.
- Baudson, T. G., and F. Preckel. 2013, January. "Teachers' Implicit Personality Theories About the Gifted: An Experimental Approach." *School Psychology Quarterly* 28, no. 1: 37–46. <https://doi.org/10.1037/spq0000011>.
- Betts, J. E., J. J. Appleton, A. L. Reschly, S. L. Christenson, and E. S. Huebner. 2010. "A Study of the Factorial Invariance of the Student Engagement Instrument (Sei): Results From Middle and High School Students." *School Psychology Quarterly* 25, no. 2: 84–93. <https://doi.org/10.1037/a0020259>.
- Bitsko, R. H., J. R. Holbrook, R. M. Ghandour, et al. 2018. "Epidemiology and Impact of Health Care Provider-Diagnosed Anxiety and Depression Among Us Children." *Journal of Developmental & Behavioral Pediatrics* 39, no. 5: 395–403. <https://doi.org/10.1097/dbp.0000000000000571>.
- Buck, K. A., and T. Dix. 2012. "Can Developmental Changes in Inhibition and Peer Relationships Explain Why Depressive Symptoms Increase in Early Adolescence?" *Journal of Youth and Adolescence* 41: 403–413. <https://doi.org/10.1007/s10964-011-9651-9>.
- Cash, T. N., and T.-J. Lin. 2022, July. "Psychological Well-Being of Intellectually and Academically Gifted Students in Self-Contained and Pull-Out Gifted Programs." *Gifted Child Quarterly* 66, no. 3: 188–207. Sage Journals. <https://doi.org/10.1177/00169862211032987>.
- Centers for Disease Control and Prevention. 2021. Youth Risk Behavior Survey: Data Summary and Trends Report. [https://www.cdc.gov/healthyouth/data/yrbs/pdf/YRBS\\_Data-Summary-Trends\\_Report2023\\_508.pdf](https://www.cdc.gov/healthyouth/data/yrbs/pdf/YRBS_Data-Summary-Trends_Report2023_508.pdf).
- Coleman, L. J. 1985. *Schooling the Gifted*. Addison-Wesley Publishing Co.
- Coleman, L. J., and T. L. Cross. 1988. "Is Being Gifted a Social Handicap?" *Journal for the Education of the Gifted* 11, no. 4: 41–56. Sage Journals. <https://doi.org/10.1177/016235328801100406>.
- Coleman, L. J., T. L. Cross, and J. R. Cross. 2021. "Lived experience, mixed messages, and stigma." In *Handbook For Counselors Serving Students With Gifts and Talents: Development, Relationships, School Issues, and Counseling Needs/interventions*, edited by T. L. Cross and J. R. Cross, 427–450. Prufrock Press.
- Coleman, L. J., K. J. Micko, and T. L. Cross. 2015, December. "Twenty-Five Years of Research on the Lived Experience of Being Gifted in School: Capturing the Students' Voices." *Journal for the Education of the Gifted* 38, no. 4: 358–376. Sage Journals. <https://doi.org/10.1177/0162353215607322>.
- Connell, J. P., and J. G. Wellborn. 1991. "Competence, Autonomy and Relatedness: A Motivational Analysis of Self-System Processes." In *Minnesota Symposium on Child Psychology*, edited by M. R. Gunnar and L. A. Sroufe, 23. Self processes in development, 43–77. University of Chicago Press.
- Crosnoe, R., M. K. Johnson, and G. H. Elder. 2004. "Intergenerational Bonding in School: The Behavioral and Contextual Correlates of Student-Teacher Relationships." *Sociology of Education* 77, no. 1: 60–81. <https://doi.org/10.1177/003804070407700103>.
- Cross, J. R., C. O'Reilly, M. Kim, S. Mammadov, and T. L. Cross. 2015. "Social Coping and Self-Concept Among Young Gifted Students in Ireland and the United States: A Cross-Cultural Study." *High Ability Studies* 26, no. 1: 39–61. <https://doi.org/10.1080/13598139.2015.1031881>.
- Cross, J. R., C. T. Vaughn, S. Mammadov, et al. 2019. "A Cross-Cultural Study of the Social Experience of Giftedness." *Roeper Review* 41, no. 4: 224–242. <https://doi.org/10.1080/02783193.2019.1661052>.
- Cross, T. 1997, June. "Psychological and Social Aspects of Educating Gifted Students." *Peabody Journal of Education* 72, no. 3: 180–200.
- Cross, T. L., L. Andersen, S. Mammadov, and J. R. Cross. 2018. "Social and Emotional Development of Students With Gifts and Talents." In *Introduction to Gifted Education*, edited by J. L. Roberts, T. F. Inman, and J. H. Robins, 77–94. Prufrock Press Inc.
- Cross, T. L., L. J. Coleman, and R. A. Stewart. 1995. "Psychosocial Diversity Among Gifted Adolescents: An Exploratory Study of Two Groups." *Roeper Review* 17, no. 3: 181–185. Taylor & Francis. <https://doi.org/10.1080/02783199509553655>.
- Cross, T. L., and R. A. Stewart. 1995, Summer. "A Phenomenological Investigation of the 'Lebenswelt' of Gifted Students in Rural High Schools." *Journal of Secondary Gifted Education* 6, no. 4: 273–280. ERIC.
- Deci, E. L., and R. M. Ryan. 2000. "The 'What' and 'Why' of Goal Pursuits: Human Needs and the Self-Determination of Behavior." *Psychological Inquiry* 11: 227–268. [https://doi.org/10.1207/S15327965PLI1104\\_01](https://doi.org/10.1207/S15327965PLI1104_01).
- Dinh, H.-V. T., C. T. Nguyen Phuoc, T.-A. T. Tran, M. L. Dau, and T. V. Nguyen. 2021. "The Vietnamese Version of the Social and Emotional Competence Questionnaire (SECQ): Psychometric Properties Among Adolescents." *TPM-Testing, Psychometrics, Methodology in Applied Psychology* 28, no. 2: 251–266.
- Faircloth, B. S., and J. V. Hamm. 2005. "Sense of Belonging Among High School Students Representing 4 Ethnic Groups." *Journal of Youth and Adolescence* 34, no. 4: 293–309. <https://doi.org/10.1007/s10964-005-5752-7>.
- Fredricks, J. A., P. C. Blumenfeld, and A. H. Paris. 2004. "School Engagement: Potential of the Concept, State of the Evidence." *Review of educational research* 74, no. 1: 59–109. <https://doi.org/10.3102/00346543074001059>.
- Fredricks, J. A., and W. McColskey. 2012. "The Measurement of Student Engagement: A Comparative Analysis of Various Methods and Student Self-Report Instruments." In *Handbook of Research on Student Engagement*, edited by S. L. Christenson, A. L. Reschly, and C. Wylie, 763–782. Springer Science + Business Media. [https://doi.org/10.1007/978-1-4614-2018-7\\_37](https://doi.org/10.1007/978-1-4614-2018-7_37).
- Furrer, C., and E. Skinner. 2003. "Sense of Relatedness as a Factor in Children's Academic Engagement and Performance." *Journal of Educational Psychology* 95, no. 1: 148–162. <https://doi.org/10.1037/0022-0663.95.1.148>.
- Geake, J. G., and M. U. M. Gross. 2008. "Teachers' Negative Affect Toward Gifted Students: An Evolutionary Psychological Study." *Gifted Child Quarterly* 52, no. 3: 217–231. <https://doi.org/10.1177/0016986208319704>.
- Georgia Department of Education. 2024. 2024-2025 Resource Manual for Gifted Education Services. [https://lor2.gadoe.org/gadoe/file/1dacad8d-a918-4dcc-afae-4ab1172eda82/1/\\_2024-2025%20Gifted%20Resource%20Manual\\_.pdf](https://lor2.gadoe.org/gadoe/file/1dacad8d-a918-4dcc-afae-4ab1172eda82/1/_2024-2025%20Gifted%20Resource%20Manual_.pdf).
- Gómez-Ortiz, O., E. M. Romera, R. Ortega-Ruiz, M. Herrera, and J. O'Higgins Norman. 2019. "Multidimensional Social Competence in Research on Bullying Involvement: A Cross-Cultural Study." *Behavioral Psychology* 27, no. 2: 217–238.
- Gómez-Ortiz, O., E. M. Romera-Félix, and R. Ortega-Ruiz. 2017. "Multidimensionality of Social Competence: Measurement of the Construct and Its Relationship With Bullying Roles." *Journal of Psychodidactics* 22, no. 1: 37–44. [https://doi.org/10.1016/S1136-1034\(17\)30042-4](https://doi.org/10.1016/S1136-1034(17)30042-4).
- Goodenow, C. 1993. "The Psychological Sense of School Membership Among Adolescents: Scale Development and Educational Correlates." *Psychology in the Schools* 30, no. 1: 79–90. [https://doi.org/10.1002/1520-6807\(199301\)30:1<79::AID-PITS2310300113>3.0.CO;2-X](https://doi.org/10.1002/1520-6807(199301)30:1<79::AID-PITS2310300113>3.0.CO;2-X).
- Gottfried, A. E., and A. W. Gottfried. 1996. "A Longitudinal Study of Academic Intrinsic Motivation in Intellectually Gifted Children: Childhood Through Early Adolescence." *Gifted Child Quarterly* 40: 179–183. <https://doi.org/10.1177/001698629604000402>.
- Gross, M. U. M. 2004. *Exceptionally Gifted Children* 2nd ed. Routledge.
- Hornstra, L., A. Bakx, S. Mathijssen, and J. J. A. Denissen. 2020. "Motivating Gifted and Non-Gifted Students in Regular Primary

- Schools: A Self-Determination Perspective." *Learning and Individual Differences* 80, Article: 101871. <https://doi.org/10.1016/j.lindif.2020.101871>.
- Hornstra, L., A. C. S. Mathijssen, J. J. A. Denissen, and A. Bakx. 2023. "Academic Motivation of Intellectually Gifted Students and Their Classmates in Regular Primary School Classes: A Multidimensional, Longitudinal, Person- and Variable-Centered Approach." *Learning and Individual Differences* 107, Article: 102345. <https://doi.org/10.1016/j.lindif.2023.102345>.
- Kanevsky, L., and T. Keighley. 2003. "To Produce or Not to Produce? Understanding Boredom and the Honor In Underachievement." *Roeper Review* 26, no. 1: 20–28. <https://doi.org/10.1080/02783190309554235>.
- Kang, D., L. M. Stough, M. Yoon, and J. Liew. 2021, January. "The Structural Association Between Teacher-Student Relationships and School Engagement: Types and Informants." *International Journal of Educational Research Open* 2: 100072. <https://doi.org/10.1016/j.ijedro.2021.100072>.
- Kitsantas, A., L. Bland, and D. S. Chirinos. 2017. "Gifted Students' Perceptions of Gifted Programs: An Inquiry Into Their Academic and Social-Emotional Functioning." *Journal for the Education of the Gifted* 40, no. 3: 266–288. Sage Journals. <https://doi.org/10.1177/0162353217717033>.
- Korpershoek, H., E. T. Canrinus, M. Fokkens-Bruinsma, and H. de Boer. 2020. "The Relationships Between School Belonging and Students' Motivational, Social-Emotional, Behavioural, and Academic Outcomes in Secondary Education: A Meta-Analytic Review." *Research Papers in Education* 35, no. 6: 641–680. <https://doi.org/10.1080/02671522.2019.1615116>.
- Košir, K., M. Horvat, U. Aram, and N. Jurinec. 2015. "Is Being Gifted Always an Advantage? Peer Relations and Self-Concept of Gifted Students." *High Ability Studies* 27, no. 2: 129–148. Taylor & Francis. <https://doi.org/10.1080/13598139.2015.1108186>.
- Lee, S.-Y., P. Olszewski-Kubilius, and D. T. Thomson. 2012. "Academically Gifted Students' Perceived Interpersonal Competence and Peer Relationships." *Gifted Child Quarterly* 56, no. 2: 90–104. Sage Journals. <https://doi.org/10.1177/0016986212442568>.
- Little, C. A. 2012. "Curriculum as Motivation for Gifted Students." *Psychology in the Schools* 49, no. 7: 695–705. <https://doi.org/10.1002/pits.21621>.
- Lovelace, M. D., A. L. Reschly, J. J. Appleton, and M. E. Lutz. 2014. "Concurrent and Predictive Validity of the Student Engagement Instrument." *Journal of Psychoeducational Assessment* 32, no. 6: 509–520. <https://doi.org/10.1177/0734282914527548>.
- Maehr, M. L., and H. A. Meyer. 1997. "Understanding Motivation and Schooling; Where We've Been, Where We Are, and Where We Need to Go." *Educational Psychology Review* 9: 371–409. <https://doi.org/10.1023/A:1024750807365>.
- Mammadov, S. 2019. "Being Gifted in Turkey: Educational and Social Experiences of High-Ability Students." *Gifted Education International* 35, no. 3: 216–236. <https://doi.org/10.1177/0261429419839397>.
- Mammadov, S. 2021. "Social Coping." In *Handbook for Counselors Serving Students With Gifts and Talents: Development, Relationships, School Issues, and Counseling Needs/interventions*, edited by T. L. Cross and J. R. Cross, 2nd ed., 801–811. Routledge.
- Mammadov, S. 2023. "Personality Profiles of Gifted Adolescents and Relations With Life Satisfaction, Perceived Social Support, and Academic Achievement." *High Ability Studies* 34, no. 1: 87–108. <https://doi.org/10.1080/13598139.2022.2068509>.
- Mammadov, S., and N. B. Hertzog. 2023. "Million-Dollar Question: What Is the Most Effective and Equitable Way to Deliver Services to Advanced Learners?" *Journal of School Leadership* 34, no. 2: 151–176. <https://doi.org/10.1177/10526846231187571>.
- Mammadov, S., and K. Schroeder. 2023. "A Meta-Analytic Review of the Relationships Between Autonomy Support and Positive Learning Outcomes." *Contemporary Educational Psychology* 75: 102235. <https://doi.org/10.1016/j.cedpsych.2023.102235>.
- McCormick, M. P., and E. E. O'Connor. 2015. "Teacher-Child Relationship Quality and Academic Achievement in Elementary School: Does Gender Matter?" *Journal of Educational Psychology* 107, no. 2: 502–516. <https://doi.org/10.1037/a0037457>.
- Mendaglio, S. 2012. "Overexcitabilities and Giftedness Research: A Call for a Paradigm Shift." *Journal for the Education of the Gifted* 35, no. 3 (September): 207–219. <https://doi.org/10.1177/0162353212451704>.
- Moon, S. M. 2009. "Myth 15: High-Ability Students Don't Face Problems and Challenges." *Gifted Child Quarterly* 53, no. 4: 274–276. <https://doi.org/10.1177/0016986209346943>.
- Mueller, C. E. 2009. "Protective Factors as Barriers to Depression in Gifted and Nongifted Adolescents." *Gifted Child Quarterly* 53, no. 1: 3–14. Sage Journals. <https://doi.org/10.1177/0016986208326552>.
- Neihart, M. 2002. "Risk and Resilience in Gifted Children: A Conceptual Framework." In *The Social and Emotional Development of Gifted Children: What Do We Know?*, edited by M. Neihart, S. Reis, N. M. Robinson, and S. M. Moon, 113–122. Prufrock Press.
- Neihart, M., Pfeiffer, S. I., and Cross, T. L., ed. 2015. *The Social and Emotional Development of Gifted Children: What Do We Know?*, (2nd ed.). Prufrock Press.
- O'Connor, E., and K. McCartney. 2007. "Examining Teacher-Child Relationships and Achievement as Part of an Ecological Model of Development." *American Educational Research Journal* 44, no. 2: 340–369. <https://doi.org/10.3102/0002831207302172>.
- Olszewski-Kubilius, P., S.-Y. Lee, and D. Thomson. 2014. "Family Environment and Social Development in Gifted Students." *Gifted Child Quarterly* 58, no. 3: 199–216. <https://doi.org/10.1177/0016986214526430>.
- Perdue, N. H., D. P. Manzeske, and D. B. Estell. 2009. "Early Predictors of School Engagement: Exploring the Role of Peer Relationships." *Psychology in the Schools* 46, no. 10: 1084–1097. <https://doi.org/10.1002/pits.20446>.
- Peterson, J., N. Duncan, and K. Canady. 2009, Winter. "A Longitudinal Study of Negative Life Events, Stress, and School Experiences of Gifted Youth." *Gifted Child Quarterly* 53, no. 1: 34–49. Sage Journals. <https://doi.org/10.1177/0016986208326553>.
- Pfeiffer, S. I. 2009, August. "The Gifted: Clinical Challenges for Child Psychiatry." *Journal of the American Academy of Child & Adolescent Psychiatry* 48, no. 8: 787–790. <https://doi.org/10.1097/CHI.0b013e3181aa039d>.
- Raufelder, D., D. Sahabandu, G. S. Martínez, and V. Escobar. 2015. "The Mediating Role of Social Relationships in the Association of Adolescents' Individual School Self-Concept and Their School Engagement, Belonging and Helplessness in School." *Educational Psychology* 35, no. 2: 137–157. <https://doi.org/10.1080/01443410.2013.849327>.
- Reschly, A. L., J. Betts, and J. J. Appleton. 2014. "An Examination of the Validity of Two Measures of Student Engagement." *International Journal of School & Educational Psychology* 2, no. 2: 106–114. <https://doi.org/10.1080/21683603.2013.876950>.
- Reschly, A. L., and S. L. Christenson. 2012. "Jingle, Jangle, and Conceptual Haziness: Evolution and Future Directions of the Engagement Construct." In *Handbook of Research on Student Engagement*, edited by S. L. Christenson, A. L. Reschly, and C. Wylie, 3–19. Springer Science + Business Media. [https://doi.org/10.1007/978-1-4614-2018-7\\_1](https://doi.org/10.1007/978-1-4614-2018-7_1).
- Reschly, A. L., E. S. Huebner, J. J. Appleton, and S. Antaramian. 2008. "Engagement as Flourishing: The Contribution of Positive Emotions and Coping to Adolescents' Engagement at School and With Learning." *Psychology in the Schools* 45, no. 5: 419–431. <https://doi.org/10.1002/pits.20306>.
- Rinn, A. N., M. J. Reynolds, and K. S. McQueen. 2011. "Perceived Social Support and the Self-Concepts of Gifted Adolescents." *Journal for the*



- Education of the Gifted* 34, no. 3: 367–396. <https://doi.org/10.1177/016235321103400302>.
- Rinn, A. N. 2018. "Social And Emotional Considerations for Gifted Students." In *APA Handbook of Giftedness and Talent*, edited by S. I. Pfeiffer, 440–460. The American Psychological Association. <https://doi.org/10.1037/0000038-000>.
- Rogers, K. B. 2007. "Lessons Learned about Educating the Gifted and Talented: A Synthesis of the Research on Educational Practice." *Gifted Child Quarterly* 51, no. 4: 382–396. <https://doi.org/10.1177/0016986207306324>.
- Roorda, D. L., H. M. Y. Koomen, J. L. Spilt, and F. J. Oort. 2011. "The Influence of Affective Teacher-Student Relationships on Students' School Engagement and Achievement: A Meta-Analytic Approach." *Review of Educational Research* 81, no. 4: 493–529. Sage Journals. <https://doi.org/10.3102/0034654311421793>.
- Rubin, K. H., W. M. Bukowski, and J. G. Parker. 2006. "Peer Interactions, Relationships, and Groups." In *Handbook of Child Psychology: Social, Emotional, and Personality Development*, edited by N. Eisenberg, W. Damon, and R. M. Lerner, 571–645. John Wiley & Sons, Inc.
- Sánchez, B., Y. Colón, and P. Esparza. 2005. "The Role of Sense of School Belonging and Gender in the Academic Adjustment of Latino Adolescents." *Journal of Youth and Adolescence* 34: 619–628. <https://doi.org/10.1007/s10964-005-8950-4>.
- Schmitt, C., and V. Goebel. 2015. "Experiences of High-Ability High School Students: A Case Study." *Journal for the Education of the Gifted* 38, no. 4: 428–446. <https://doi.org/10.1177/0162353215607325>.
- Seligman, M. E. P. 2011. *Flourish: A New Understanding of Happiness and Well-being and How to Achieve Them*. Nicholas Brealey Publishing.
- Shechtman, Z., and A. Silektor. 2012. "Social Competencies and Difficulties of Gifted Children Compared to Nongifted Peers." *Roeper Review* 34, no. 1: 63–72. <https://doi.org/10.1080/02783193.2012.627555>.
- Silverman, L. K. 2012. "Asynchronous Development: A Key to Counseling the Gifted." In *Handbook for Counselors Serving Students With Gifts and Talents*, edited by T. L. Cross and J. R. Cross, 261–279. Prufrock Press.
- Skinner, E., C. Furrer, G. Marchand, and T. Kindermann. 2008. "Engagement and Disaffection in the Classroom: Part of a Larger Motivational Dynamic?" *Journal of Educational Psychology* 100, no. 4: 765–781. <https://doi.org/10.1037/a0012840>.
- Skinner, E. A., T. A. Kindermann, and C. J. Furrer. 2009. "A Motivational Perspective on Engagement and Disaffection: Conceptualization and Assessment of Children's Behavioral and Emotional Participation in Academic Activities in the Classroom." *Educational and Psychological Measurement* 69, no. 3: 493–525. <https://doi.org/10.1177/0013164408323233>.
- Tannenbaum, A. J. 1983. *Gifted Children: Psychological and Educational Perspectives*. Macmillan.
- Twardowska-Staszek, E., and I. Zych. 2023. "Social and Emotional Competencies of Polish Pupils: Psychometric Properties of the Polish Version of the Social and Emotional Competencies Questionnaire (Sec-Q)." *Multidisciplinary Journal of School Education* 12, no. 24: 187–202. <https://doi.org/10.35765/mjse.2023.1224.09>.
- Vogl, K., and F. Preckel. 2014. "Full-Time Ability Grouping of Gifted Students: Impacts on Social Self-Concept and School-Related Attitudes." *Gifted Child Quarterly* 58, no. 1: 51–68. Sage Journals. <https://doi.org/10.1177/0016986213513795>.
- Williams, K. D., and J. Gerber. 2005, January. "Making Minds Ii: The Making of the Ignored and Excluded Mind." *Interaction Studies. Social Behaviour and Communication in Biological and Artificial Systems* 6, no. 3: 359–374. <https://doi.org/10.1075/is.6.3.04wil>.
- Wirthwein, L., S. Bergold, F. Preckel, and R. Steinmayr. 2019. "Personality and School Functioning of Intellectually Gifted and Nongifted Adolescents: Self-Perceptions and Parents' Assessments." *Learning and Individual Differences* 73: 16–29. <https://doi.org/10.1016/j.lindif.2019.04.003>.
- Wong, Z. Y., and G. A. D. Liem. 2022. "Student Engagement: Current State of the Construct, Conceptual Refinement, and Future Research Directions." *Educational Psychology Review* 34: 107–138. <https://doi.org/10.1007/s10648-021-09628-3>.
- Zee, M., H. M. Y. Koomen, and I. Van der Veen. 2013. "Student-Teacher Relationship Quality and Academic Adjustment in Upper Elementary School: The Role of Student Personality." *Journal of School Psychology* 51, no. 4: 517–533. <https://doi.org/10.1016/j.jsp.2013.05.003>.
- Zych, I., R. Ortega-Ruiz, R. Muñoz-Morales, and V. J. Llorent. 2018. "Dimensions and Psychometric Properties of the Social and Emotional Competencies Questionnaire (Sec-Q) in Youth and Adolescents." *Revista Latinoamericana de Psicología* 50, no. 2: 98–106. <https://doi.org/10.14349/rfp.2018.v50.n2.3>.